Answers

Worksheet

Rounding integers to one significant figure

Task A: Identifying significant figures

Circle the first **significant figure** in each of the following:

a) 150 267

b) 3560 f)4 987 932

c) 60 253 461

d) 137

e) 98 200

f) 4 987 932

Task B: Using a number line for significant figures

1) Match the following to the correct rounded to one significant figure number:

3 487 925	thirty
28 021	three hundred
28	thirty thousand
348.7925	three million

2) Complete the following:

Number	The 1st significant figure is in the _______ column.	Round to the nearest _______.	Number line	To 1 s.f.
36	tens	10	30 35 40	40
832	hundreds	100	800 850 900	800
3519	thousands	1000	3000 3500 4000	4000
43 867.09	10 thousands	10 000	40 000 45 000 50 000	40 000
992 782	100 thousands	100 000	900 000 950 000 1 000 000	1 000 0000

Task C: Rounding to one significant figure

1) Write each of the numbers below to one significant figure, find the answer in the grid to complete this joke:

A sheepdog goes out to get the farmer's sheep in. He comes back and says to the farmer, 'that's all 100 sheep in the barn' The farmer says, 'that's odd we only had 97 this morning'. The sheepdog replies …

A	B	C	D	E	F	G	H	I	J	K	L	M
200	60	400	30 000	10	20 000	700	90	5000	20	100	9000	80 000
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
70	2000	7000	40	8000	300	10 000	800	900	1000	50 000	40	600

1) 4682 ≈ *5000* *I*
2) 1800 ≈ *2000* *O*
3) 1499 ≈ *1000* *W*
4) 71 ≈ *70* *N*
5) 104 ≈ *100* *K*
6) 5498 ≈ *5000* *I*

7) 65 ≈ *70* *N*
8) 13 ≈ *10* *E*
9) 840 ≈ *800* *U*
10) 8287 ≈ *8000* *R*
11) 34 999 ≈ *30 000* *D*
12) 1560 ≈ *2000* *O*
13) 25 873 ≈ *30 000* *D*

14) 75 001 ≈ *80 000* *M*
15) 94 ≈ *90* *H*
16) 9810 ≈ *10 000* *T*
17) 13 ≈ *10* *E*
18) 7459 ≈ *7000* *P*
19) 762 ≈ *800* *U*

'I know, I rounded them up!'

2) What values can the missing digit take?

a) 4?90 = 4000 (1 s.f.)

0, 1, 2, 3 or 4

b) 9?12 = 10 000 (1 s.f.)

5, 6, 7, 8, or 9

c) 34 ?90 = 30 000 (1 s.f.)

any digit

d) 678? = 7000 (1 s.f.)

any digit